

# PROJECT REPORT

## Meeting Room Booking System

*using ServiceNow App Engine Studio*

|               |                                                                |
|---------------|----------------------------------------------------------------|
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| Project Title | Meeting Room Booking System using ServiceNow App Engine Studio |
| Platform      | ServiceNow App Engine Studio                                   |
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# 1. INTRODUCTION

## 1.1 Project Overview

The Meeting Room Booking System is a custom application developed using ServiceNow App Engine Studio. It digitalizes the meeting room reservation process within an organization, replacing manual methods such as emails, spreadsheets, and phone calls with an automated, centralized platform.

Employees can submit booking requests through an intuitive online form by entering their details, selecting a room, and specifying the meeting date, time, and purpose. The system automatically validates the information, assigns booking statuses, sends email notifications, and stores all records in a centralized database.

|                   |                                                |
|-------------------|------------------------------------------------|
| Application Name  | Meeting Room Booking System                    |
| Platform          | ServiceNow App Engine Studio                   |
| Custom Table      | Meeting Room Booking (x_mrbs_booking)          |
| Booking ID Prefix | MRB (e.g. MRB0001001)                          |
| User Roles        | Employee (requestor), Administrator (approver) |

## 1.2 Purpose

The primary purpose of this project is to automate and simplify the meeting room booking process by replacing manual, error-prone methods with a digital ServiceNow application. The system aims to:

- Eliminate double-bookings and scheduling conflicts through validation logic.
- Provide a single, centralized platform for employees to raise and track booking requests.
- Automate status management and notifications using Business Rules, Flow Designer, and Email Notifications.
- Give administrators real-time visibility into all bookings through dashboards and reports.
- Maintain a complete booking history for audit, analysis, and future planning.

## 2. IDEATION PHASE

### 2.1 Problem Statement

In many organizations, meeting rooms are booked through informal, manual processes — sending emails, making phone calls, or maintaining shared spreadsheets. This leads to several recurring problems:

| P<br>S           | Problem                 | Impact                                                                            |
|------------------|-------------------------|-----------------------------------------------------------------------------------|
| P<br>S<br>-<br>1 | Double Booking          | Two teams occupy the same room at the same time, disrupting both meetings.        |
| P<br>S<br>-<br>2 | No Real-Time Visibility | Employees cannot see which rooms are free without calling or emailing the admin.  |
| P<br>S<br>-<br>3 | Human Errors            | Manual data entry in spreadsheets leads to incorrect dates, times, or room names. |
| P<br>S<br>-<br>4 | No Booking History      | There is no central record of past bookings for audit or analysis.                |
| P<br>S<br>-<br>5 | Approval Delays         | Booking approvals depend on manual communication, causing delays.                 |

### 2.2 Empathy Map Canvas

An empathy map was used to understand the perspective, frustrations, and goals of the primary user — the employee who needs to book a meeting room.

| THINK & FEEL                                                                                                                                                                                     | SEE                                                                                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Wants instant confirmation</li><li>• Worried about double-booking</li><li>• Frustrated by manual process</li></ul>                                       | <ul style="list-style-type: none"><li>• Shared spreadsheet with no real-time view</li><li>• Rooms listed as occupied unexpectedly</li><li>• Long email threads to confirm availability</li></ul>          |
| HEAR                                                                                                                                                                                             | SAY & DO                                                                                                                                                                                                  |
| <ul style="list-style-type: none"><li>• Colleagues complaining about room clashes</li><li>• Manager asking why rooms weren't ready</li><li>• IT team saying 'raise a request and wait'</li></ul> | <ul style="list-style-type: none"><li>• Emails or calls admin to check availability</li><li>• Physically visits the room before meeting</li><li>• Re-confirms the booking close to meeting time</li></ul> |
| PAIN                                                                                                                                                                                             | GAIN                                                                                                                                                                                                      |

|                                                                                                                                                                 |                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Double booking conflicts</li><li>• No real-time room availability</li><li>• Manual errors and approval delays</li></ul> | <ul style="list-style-type: none"><li>• Instant self-service booking</li><li>• Automatic confirmation email</li><li>• Centralized booking history</li></ul> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|

2.3 Brainstorming

The team conducted a brainstorming session to identify ideas addressing the core booking problem, which were then grouped and prioritized by importance and feasibility.

| Group                 | Ideas                                                                        | Priority     |
|-----------------------|------------------------------------------------------------------------------|--------------|
| Booking & Validation  | Online booking form, room selection, time validation, auto-number ID         | Must Have    |
| Automation & Workflow | Business Rule for default status, Flow Designer, admin approval workflow     | Must Have    |
| Notifications         | Email notification on booking creation and status change                     | Must Have    |
| Reporting & Analytics | Dashboard with status chart, room-wise chart, booking trend and list reports | Should Have  |
| Future Enhancements   | Recurring bookings, QR check-in, calendar integration, mobile app            | Future Scope |

### 3. REQUIREMENT ANALYSIS

#### 3.1 Customer Journey Map

The Customer Journey Map traces the complete experience of an employee and administrator from discovering the need for a meeting room to completing the booking and reviewing analytics.

| Phase                    | Key Actions & Goals                                                                                            |
|--------------------------|----------------------------------------------------------------------------------------------------------------|
| 1. Discover              | Employee recognizes need for a room → Discovers the ServiceNow Booking App via portal                          |
| 2. Access App            | Logs in via SSO → Opens Meeting Room Booking app → Views available rooms on form                               |
| 3. Submit Booking        | Fills form (Name, ID, Email, Room, Date, Start Time, End Time, Purpose) → Clicks Submit                        |
| 4. Booking Management    | System validates, assigns Pending status, triggers Flow, sends email notification → Admin reviews and approves |
| 5. Completion & Feedback | Meeting conducted → Admin marks Completed → Dashboard and Reports updated automatically                        |

#### 3.2 Solution Requirements

The solution requirements are divided into Functional Requirements (what the system must do) and Non-Functional Requirements (how the system must perform).

**Functional Requirements:**

| FR No. | Epic                      | Key Requirements                                                            |
|--------|---------------------------|-----------------------------------------------------------------------------|
| FR-1   | User Authentication       | SSO login, role-based access (Employee / Administrator)                     |
| FR-2   | Meeting Room Booking      | Online form with Room Name, Date, Start/End Time, Purpose, Employee details |
| FR-3   | Booking Validation        | Client Script validates End Time > Start Time; mandatory field validation   |
| FR-4   | Status Management         | Auto Pending on create; Admin changes to Approved / Completed               |
| FR-5   | Automation & Notification | Flow Designer triggers on insert; email notification sent automatically     |
| FR-6   | Reports & Dashboard       | Status chart, room-wise chart, booking trend chart, booking list report     |

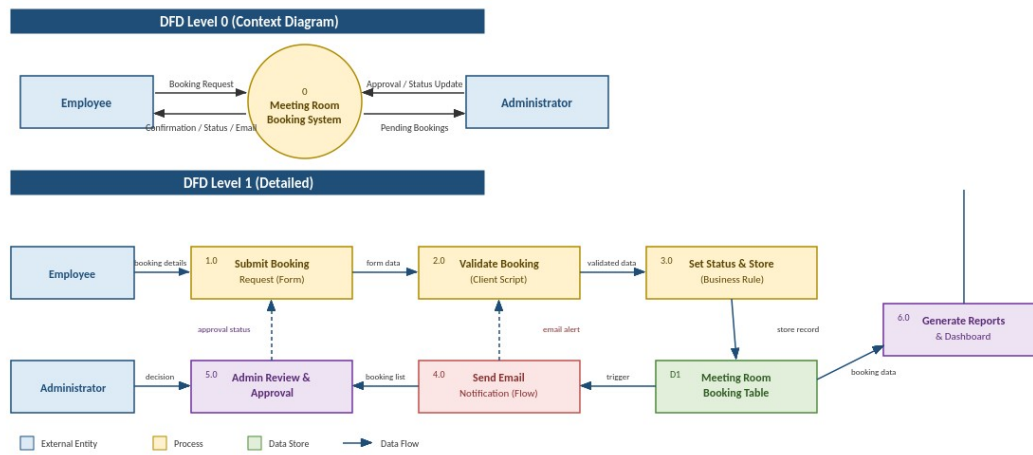
**Non-Functional Requirements:**

| NFR No. | Requirement  | Description                                                                |
|---------|--------------|----------------------------------------------------------------------------|
| NFR-1   | Usability    | Simple intuitive form; employees can book in a few clicks without training |
| NFR-2   | Security     | Role-based ACLs; only authenticated users can create/approve bookings      |
| NFR-3   | Reliability  | Every booking record reliably saved; validation rules applied consistently |
| NFR-4   | Performance  | Form submission and updates complete within 2-3 seconds under normal load  |
| NFR-5   | Availability | Available during business hours leveraging ServiceNow's platform uptime    |

|              |                        |                                                                            |
|--------------|------------------------|----------------------------------------------------------------------------|
| <b>NFR-6</b> | <b>Scalability</b>     | Supports increasing rooms, employees and bookings without performance loss |
| <b>NFR-7</b> | <b>Maintainability</b> | Modular Business Rules and Flows; easy to extend with new rooms or fields  |

### 3.3 Data Flow Diagram

The Data Flow Diagram (DFD) illustrates how data moves through the Meeting Room Booking System from booking submission to storage, notification, approval, and reporting.



The DFD is presented at two levels:

- Level 0 (Context Diagram): Shows the system as a single process with data flows to and from Employee and Administrator external entities.
- Level 1 (Detailed): Breaks the system into 6 sub-processes — Submit Booking, Validate Booking, Set Status & Store, Send Email Notification, Admin Review & Approval, and Generate Reports & Dashboard.



### 3.4 Technology Stack

The Meeting Room Booking System is built entirely on the ServiceNow Now Platform. The table below summarizes all components and technologies used.

| S.No | Component           | Description                         | Technology                                      |
|------|---------------------|-------------------------------------|-------------------------------------------------|
| 1.   | User Interface      | Employee/admin interaction layer    | App Engine Studio UI, JavaScript (Now Platform) |
| 2.   | Application Logic-1 | Client-side booking validation      | Client Script (JavaScript)                      |
| 3.   | Application Logic-2 | Auto-assigns default booking status | Business Rule (Server-side JavaScript)          |
| 4.   | Application Logic-3 | Automates the booking workflow      | Flow Designer (ServiceNow native)               |
| 5.   | Database            | Stores all booking records          | ServiceNow Custom Table (Platform DB)           |
| 6.   | Notification        | Sends email on booking creation     | ServiceNow Email Notification (SMTP)            |
| 7.   | Reporting           | Dashboards and analytical reports   | ServiceNow Reports & Dashboard                  |
| 8.   | Infrastructure      | Deployment environment              | ServiceNow Cloud (multi-instance)               |

## 4. PROJECT DESIGN

### 4.1 Problem Solution Fit

The Problem-Solution Fit canvas maps the core problems identified during the ideation phase to the specific features of the ServiceNow-based solution.

| PROBLEM                                | SOLUTION                                                            |
|----------------------------------------|---------------------------------------------------------------------|
| • Double booking of the same room/time | • Client Script validates time slots before submission              |
| • No centralized booking system        | • ServiceNow custom table as single source of truth                 |
| • Manual approval delays               | • Business Rule auto-sets Pending; Flow Designer automates workflow |
| • No confirmation to employee          | • Email Notification sent automatically on booking creation         |
| • No booking history or reporting      | • Dashboard with charts and detailed booking list reports           |
| • Lack of real-time visibility         | • Dashboard updated in real-time from the central booking table     |

### 4.2 Proposed Solution

The proposed solution is a ServiceNow application developed using App Engine Studio that enables employees to digitally book meeting rooms and administrators to manage those bookings efficiently.

Key components of the proposed solution:

- **Booking Form:** A guided online form with mandatory fields — Employee Name, Employee ID, Employee Email, Room Name, Booking Date, Start Time, End Time, and Purpose.
- **Client Script:** Validates that the End Time is always greater than the Start Time, preventing invalid submissions.
- **Business Rule:** Automatically sets the booking Status to Pending whenever a new booking record is created.
- **Flow Designer:** Triggered on record insertion to execute the booking workflow automatically.
- **Email Notification:** Sends a booking confirmation to the employee with full booking details.
- **Dashboard & Reports:** Provides total booking count, status chart (Pending/Approved/Completed), room-wise booking chart, and trend chart.

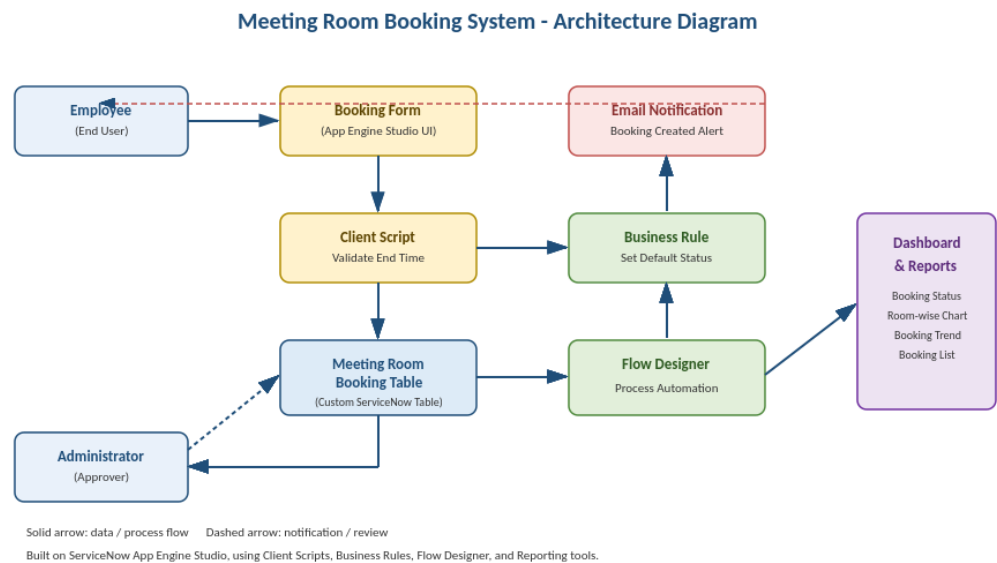
#### Booking Workflow:

| Step | Action                                                   |
|------|----------------------------------------------------------|
| 1    | Employee fills and submits the booking form              |
| 2    | Client Script validates End Time > Start Time            |
| 3    | Business Rule sets Status = Pending                      |
| 4    | Flow Designer executes and triggers notifications        |
| 5    | Email Notification sent to employee with booking details |

|           |                                                         |
|-----------|---------------------------------------------------------|
| <b>6</b>  | Booking record stored in the Meeting Room Booking Table |
| <b>7</b>  | Administrator reviews booking from the admin worklist   |
| <b>8</b>  | Administrator changes Status to Approved                |
| <b>9</b>  | Employee attends the meeting in the confirmed room      |
| <b>10</b> | Administrator marks Status as Completed                 |
| <b>11</b> | Dashboard and Reports updated automatically             |

### 4.3 Solution Architecture

The architecture diagram below illustrates the end-to-end flow of the Meeting Room Booking System, showing how the Employee, Booking Form, Client Script, Business Rule, Meeting Room Booking Table, Flow Designer, Email Notification, Administrator, and Dashboard are interconnected.



*The architecture is built entirely within the ServiceNow platform, ensuring tight integration, security, and maintainability without any external dependencies.*

## 5. PROJECT PLANNING & SCHEDULING

### 5.1 Project Planning

The project was executed in 7 structured phases (sprints), each with clear deliverables and timelines.

| Phase | Sprint                | Deliverables                                                                | Duration | Status      |
|-------|-----------------------|-----------------------------------------------------------------------------|----------|-------------|
| 1     | Application Setup     | Created Meeting Room Booking System app; configured roles (Admin/User)      | Week 1   | ✓ Completed |
| 2     | Database Design       | Custom table, auto-numbering (MRB prefix), all required fields              | Week 1-2 | ✓ Completed |
| 3     | Form Design           | Booking form, mandatory fields, choice values for Room and Status           | Week 2   | ✓ Completed |
| 4     | Business Logic        | Client Script (End Time validation), Business Rule (default Pending status) | Week 3   | ✓ Completed |
| 5     | Workflow Automation   | Flow Designer flow, Email Notification configuration                        | Week 3-4 | ✓ Completed |
| 6     | Reporting & Analytics | Reports (status, room-wise, trend, list), Dashboard with 5 widgets          | Week 4   | ✓ Completed |
| 7     | Testing & Validation  | Booking creation tests, Client Script, Business Rule, Flow, Notification    | Week 5   | ✓ Completed |

## 6. FUNCTIONAL AND PERFORMANCE TESTING

### 6.1 Performance Testing

The Meeting Room Booking System was tested across all functional components to verify correctness, automation, and notification behaviour. The following test cases were executed:

| TC No. | Module             | Test Case                                   | Expected Result                                   | Priority | Result |
|--------|--------------------|---------------------------------------------|---------------------------------------------------|----------|--------|
| TC-01  | Booking Form       | Submit a booking with all valid fields      | Booking record created with auto-generated ID     | High     | ✓ Pass |
| TC-02  | Booking Form       | Submit booking with mandatory field missing | Form shows validation error, blocks submission    | High     | ✓ Pass |
| TC-03  | Client Script      | Enter End Time earlier than Start Time      | Error message displayed; submission blocked       | High     | ✓ Pass |
| TC-04  | Client Script      | Enter End Time equal to Start Time          | Error message displayed; submission blocked       | High     | ✓ Pass |
| TC-05  | Business Rule      | Create a new booking record                 | Status auto-assigned as Pending                   | High     | ✓ Pass |
| TC-06  | Flow Designer      | Submit a new booking request                | Flow Designer executes successfully on insert     | High     | ✓ Pass |
| TC-07  | Email Notification | Submit a new booking request                | Confirmation email received by employee           | High     | ✓ Pass |
| TC-08  | Auto Number        | Submit a new booking request                | Booking ID generated in format MRB0001001         | High     | ✓ Pass |
| TC-09  | Admin Approval     | Admin opens pending booking                 | Admin can update Status to Approved               | High     | ✓ Pass |
| TC-10  | Completion         | Admin marks booking Completed               | Status updated to Completed in booking record     | Medium   | ✓ Pass |
| TC-11  | Dashboard          | Open Reports & Dashboard                    | All charts and widgets render correctly           | Medium   | ✓ Pass |
| TC-12  | Reports            | Generate Booking Status Report              | Report displays accurate booking counts by status | Medium   | ✓ Pass |

All 12 test cases passed successfully. The system performed consistently within the expected 2–3 second response time for form submission, validation, and status updates under normal load.

## 7. RESULTS

### 7.1 Output Screenshots

The Meeting Room Booking System was successfully implemented on the ServiceNow platform. The following outputs were verified during testing and demonstration:

| S.No | Screen / Output                  | Description                                                                                                                             |
|------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 1.   | Application Homepage             | Meeting Room Booking System app visible on ServiceNow portal after App Engine Studio setup.                                             |
| 2.   | Booking Form                     | All fields visible — Employee Name, ID, Email, Room Name (choice), Booking Date, Start Time, End Time, Purpose, and Status.             |
| 3.   | Validation Error (Client Script) | Red error message displayed when End Time is set earlier than or equal to Start Time, blocking submission.                              |
| 4.   | Successful Booking Record        | Booking record created with auto-generated ID (e.g. MRB0001001) and Status automatically set to Pending.                                |
| 5.   | Email Confirmation               | Email notification received by employee with booking details immediately after submission.                                              |
| 6.   | Admin Booking List               | Administrator sees all submitted bookings in a filterable list with status indicators.                                                  |
| 7.   | Approved Booking                 | Administrator successfully changes Status from Pending to Approved; change reflected in the booking record.                             |
| 8.   | Completed Booking                | After the meeting, administrator marks the booking Status as Completed.                                                                 |
| 9.   | Dashboard                        | Dashboard displays Total Bookings, Booking Status Chart (Pending/Approved/Completed), Room-wise Booking Chart, and Booking Trend Chart. |
| 10.  | Reports                          | Booking Status Report, Room-wise Report, Trend Report, and Booking List Report generated successfully.                                  |

## 8. ADVANTAGES & DISADVANTAGES

| ADVANTAGES                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | DISADVANTAGES / LIMITATIONS                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>✓ Eliminates double-booking through time validation</li><li>✓ Centralized, real-time booking management</li><li>✓ Automatic email confirmation reduces manual communication</li><li>✓ Role-based access ensures data security</li><li>✓ Dashboard provides instant visibility into room utilization</li><li>✓ No additional cost — built on existing ServiceNow platform</li><li>✓ Auto-generated booking IDs enable easy tracking</li></ul> | <ul style="list-style-type: none"><li>✗ Requires ServiceNow platform license</li><li>✗ No real-time room occupancy sensor integration</li><li>✗ Mobile access depends on ServiceNow mobile app availability</li><li>✗ No built-in recurring booking feature (future scope)</li><li>✗ Admin approval is manual; no automated approval for low-risk bookings</li><li>✗ No conflict detection across overlapping bookings currently</li></ul> |



## 9. CONCLUSION

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The Meeting Room Booking System, developed using ServiceNow App Engine Studio, successfully achieves its primary objective of automating and centralizing the meeting room reservation process within an organization.

The application replaces manual, error-prone booking methods — emails, spreadsheets, and phone calls — with a structured digital workflow that validates bookings, automates status management, sends instant confirmations, and provides complete visibility through dashboards and reports.

All seven development phases were completed as planned: application setup, database design, form design, business logic implementation, workflow automation, reporting/analytics, and testing. All 12 test cases passed successfully, validating the correctness and reliability of the system.

The project demonstrates the effective use of ServiceNow's low-code App Engine Studio capabilities, including:

- Custom Table with Auto-Numbering for unique booking identification.
- Client Script for real-time form validation.
- Business Rule for automatic status management.
- Flow Designer for end-to-end process automation.
- Email Notification for instant booking confirmation.
- Dashboard and Reports for data-driven decision-making.

The system improves meeting room utilization, eliminates booking conflicts, and reduces the administrative burden on both employees and administrators — making it a practical and scalable solution for any organization already using the ServiceNow platform.

## 10. FUTURE SCOPE

The Meeting Room Booking System can be enhanced with the following features in future development iterations:

| S.No | Enhancement                   | Description                                                                                                          |
|------|-------------------------------|----------------------------------------------------------------------------------------------------------------------|
| 1.   | Real-Time Room Availability   | Display live room occupancy so employees can see available slots without submitting a form.                          |
| 2.   | Automatic Conflict Detection  | Prevent bookings that overlap with existing approved reservations for the same room.                                 |
| 3.   | Recurring Bookings            | Allow employees to book a room for a weekly or monthly recurring series.                                             |
| 4.   | Calendar Integration          | Sync bookings with Microsoft Outlook and Google Calendar via API integration.                                        |
| 5.   | Mobile Application            | Provide a mobile-friendly interface for employees to book and track rooms from any device.                           |
| 6.   | QR Code Check-In              | Enable employees to check into a booked room by scanning a QR code, automatically updating the booking record.       |
| 7.   | AI-Based Room Recommendations | Use ServiceNow Predictive Intelligence to suggest the most suitable room based on meeting size and past preferences. |
| 8.   | Manager Approval Workflow     | Add an optional manager approval step for bookings exceeding a defined duration.                                     |
| 9.   | Automated Room Release        | Automatically release a room if the employee does not check in within a defined period after the meeting start time. |
| 10.  | Advanced Analytics            | Provide peak-hour analysis, room utilization scores, and under-utilized room reports.                                |

## 11. APPENDIX

---

The following resources are associated with the Meeting Room Booking System project:

| Resource              | Details                                                                |
|-----------------------|------------------------------------------------------------------------|
| Platform              | ServiceNow Developer Instance (Personal Developer Instance — PDI)      |
| Application Scope     | x_mrbs_meeting_room_booking_system (scoped app)                        |
| Custom Table Name     | Meeting Room Booking (x_mrbs_booking)                                  |
| Auto Number Format    | MRB0001001, MRB0001002, MRB0001003 ...                                 |
| Source Code           | Available in ServiceNow App Engine Studio under the application scope. |
| GitHub / Project Demo | [To be added by student]                                               |
| Dataset / Reference   | [To be added by student]                                               |

### Technologies Referenced:

- ServiceNow App Engine Studio: <https://developer.servicenow.com>
- ServiceNow Flow Designer Documentation: <https://docs.servicenow.com>
- ServiceNow Business Rules: <https://docs.servicenow.com/bundle/tokyo-application-development/page/script/business-rules>
- Customer Journey Mapping: <https://www.mural.co/templates/customer-journey-map>
- DFD Reference: <https://www.ibm.com/cloud/architecture>